BIOCHEMISTRY CUMULATIVE EXAMINATION
APRIL 20, 2001

1. (40 pts) Although the double-reciprocal plot is the most widely used plotting form for enzyme kinetic data, it suffers from a major disadvantage. If linear increments of substrate concentration are used, thereby minimizing measurement errors in the laboratory, data points will be obtained that cluster near the vertical axis. Thus the intercept on the ordinate can be determined with great accuracy, but the slope of the line will be subject to considerable error, because the least reliable data points, those obtained at low substrate concentrations, have greater weight in establishing the slope. Because of the limitation of double-reciprocal plots described above, other linear plotting forms have been devised. One of these, the Hanes-Woolf plot, ([S]/V versus [S]) is perhaps the most useful in minimizing the difficulties of the double-reciprocal plot.
   a) Rearrange the Michaelis-Menten equation to give [S]/V as a function of [S].
   b) What is the significance of the slope, the vertical intercept, and the horizontal intercept in a plot of [S]/V versus [S].

2. (30pts) Draw the structure of the peptide His-Cys-Ser at pH 7. Show each atom and each bond of His-Cys-Ser. Pick one set of 6 atoms that lie in a plane and circle each atom.

3 (30pts) A glutamine residue that is the amino-terminal residue of a peptide often undergoes spontaneous cyclization to form a heterocyclic ring; the cyclization is accompanied by the release of ammonium ion. Diagram the structure of the ring, showing it linked to an adjacent amino acid residue.